Navigating abstract semantic domains in a multilingual environment

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Background

- Languages map words to concepts in distinct ways in many domains like color, space, and kinship[1,2,3]
- For example, Hindi has highly informative kinship terms that may specify features of relationships like lineage or seniority, which may lead to extended acquisition trajectories
- Speaking a language that highlights these features may also lead to effects of *semantic accenting,* causing an individual to choose different (i.e., more informative) words to describe a relationship in their L2 than a native speaker would[4]

The current studies investigate:

1. The acquisition trajectory for various terms in the highly complex & informative Hindi kinship system [**Study 1**]
2. Whether, when asked to refer in English to individuals who differ along dimensions such as lineage or seniority, multilingual Hindi native-speakers will more frequently use modifiers than English native-speakers – i.e., have a *semantic accent* [**Study 2**]
3. Whether any effect of semantic accenting varies as a function of exposure to English, i.e., whether Hindi native-speakers who are more fluent in English produce fewer modifiers in our task than those who are less fluent in English [**Study 2**]

Study 1 Methods

Participants

- N = 194 (51% female), M_{age} = 10;2
- Recruited from an English-medium school in Vadodara, Gujarat, India

Age group (grade)	*N*	Question types included
KG/1st	49	Free response only
3rd	49	Both
6th	51	Both
9th	49	Both

Procedure

Partial Knowledge Question Types

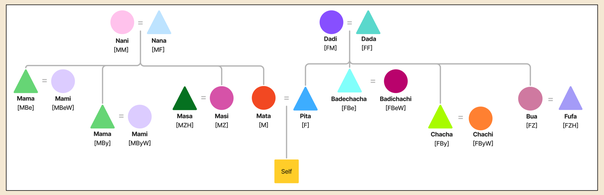
1. **Free Response**

   "How do you know when someone is a *masi*?"
   → **At what age do children have full/precise knowledge of each kinship term?**
   → **Is knowledge of the kinship term predicted by its semantic complexity?**

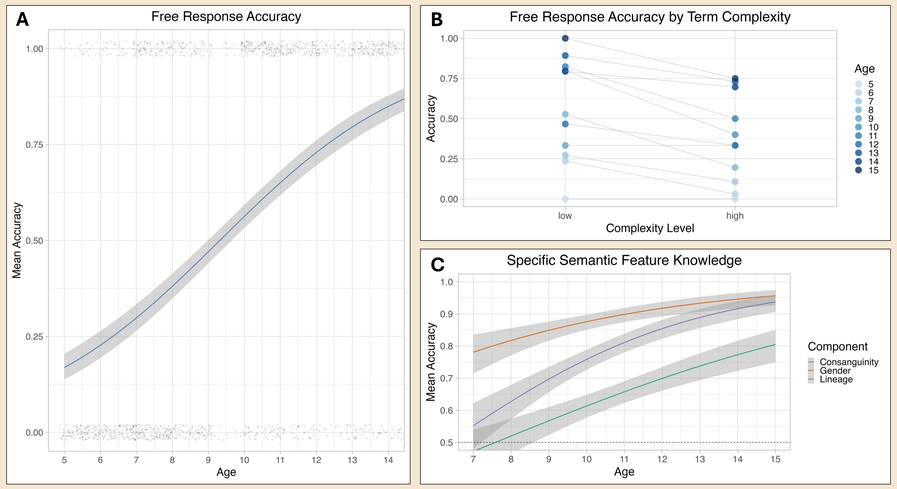
2. **Feature-Specific**

   "Is a *masi*: a) a man or a woman? [gender]
   b) on your mom's side or your dad's side? [lineage]
   c) related to you by blood or marriage?" [consanguinity]
   → **Is knowledge of some semantic features (i.e., those that are more perceptually available) incorporated into children's representations earlier than others?**

Hindi Kinship Terms (same color = same label)



Study 1 Results



A. **The acquisition trajectory of Hindi kinship terms is long, and participants were not at ceiling on a free response measure by age 15**
B. **More semantically complex terms (those that encode more features) are more difficult to learn on average than less semantically complex terms**
C. **Semantic features that are more perceptually (and morphologically) available, like gender, are often learned earlier than those that are less available, like consanguinity (blood vs. marital relation)**

Study 2 Methods

Participants

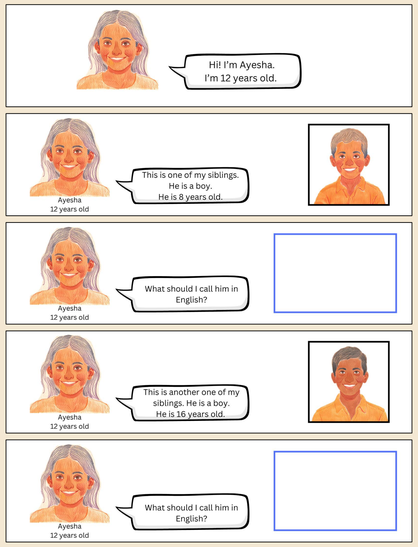
Sample	Hindi (child)	English (child)	Hindi (adult)	English (adult)
Size	*N* = 74	*N* = 12/72	*N* = 72	*N* = 72
Prediction	high modifier use	little/no modifier use	high modifier use	little/no modifier use
Mean age	14;2	14;2		

Procedure

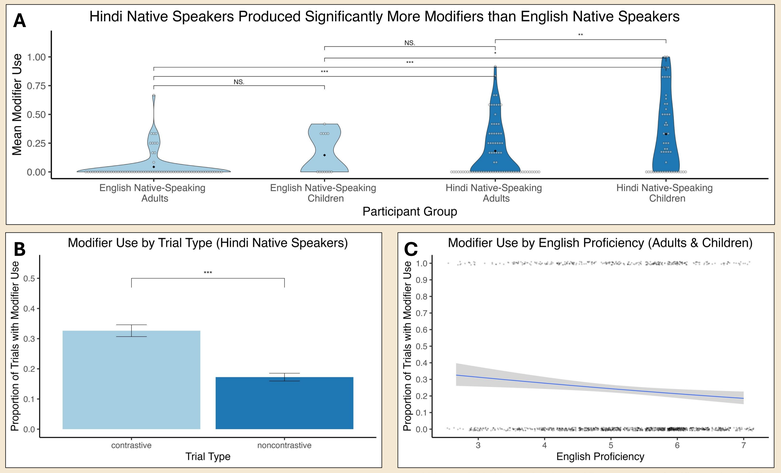
1. **Naming Task**
   Assesses modifier use for contrastive and noncontrastive term pairs – those with and without distinct labels in Hindi
2. **Modifier Test**
   Confirms participants' knowledge of modifiers expected in the naming task
3. **Background Questionnaire**
   Gauges participants' familiarity with English, Hindi, and other languages they speak, and sibling status (which may influence modifier use)

Naming Task Example

contrastive example:
younger/older brother (below)
noncontrastive example:
younger/older maternal uncle



Study 2 Results



*** p < .001, ** p < .01, * p < .05, n.s. = not significant (p > .05)

A. **English native-speakers used significantly fewer modifiers than Hindi native-speakers did**
B. **Modifier use by Hindi native-speakers was higher on contrastive than non-contrastive trials**
C. **Hindi native-speakers with high levels of English proficiency were slightly less likely to produce modifiers in the naming task than those with lower proficiency**

Discussion

- **[Study 1]** Acquisition is predicted by semantic complexity of the specific term, and certain semantic features are incorporated earlier than others depending on perceptual/morphological salience
- **[Study 2]** Preliminary evidence of a semantic accent in this domain, perhaps a result of Hindi speakers having acquired highly informative Hindi kinship terms and the features they encode
- **Together, these results reveal how individuals learn and use words in an abstract semantic domain, when the domain's categorization structure is different in their L1 and L2**
- Next steps include:
  - Collecting a full sample of English native-speaking children
  - This could either be a language-specific effect, or a general bias of L2 speakers towards informativeness when speaking English
    → Follow-up study with Spanish native speakers underway, whose kinship system is analogous to the English one

References & Acknowledgements

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3 Berlin, B., & Kay, P. (1991). Basic color terms: Their universality and evolution. Univ of California Press.
4 Malt, B. C., & Yang, X. (2024). Dual drivers of bilingual semantic accent: Semantic relations and input limitations. International Journal of Bilingualism.

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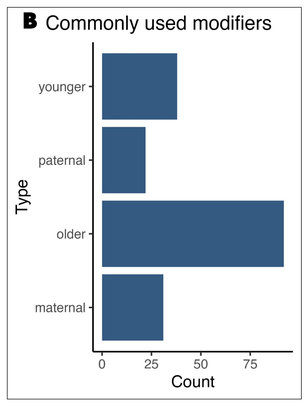
5 Fairchild, S., Mathis, A., & Papafragou, A. (2020). Pragmatics and social meaning: Understanding under-informativeness in native and non-native speakers. Cognition.



Language Questionnaire

1. On a scale of 1-7, how fluent are you in English?

1 2 3 4 5 6 7
no fluency some fluency native fluency

2. On a scale of 1-7, how often do you speak English in your daily life?

1 2 3 4 5 6 7
never a few times per week many times per day

3. Think about talking to someone who knows both Hindi and English. How often would you talk to them in English?

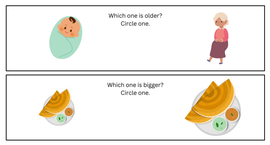
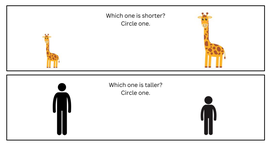
1 2 3 4 5 6 7
never choose English either language always choose English

4. When did you start learning English? Write the age.

Modifier Test

- Two alternative forced-choice design with images
- 6 trials, 4 targets and 2 fillers

Example Targets



Example Fillers